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Inventor(s)	Tillman-Davis; Michael William

Rapid release rooftop tent mount

Abstract

A bottom bracket made with a box at its center and a flange on opposing sides. Over each flange, on the side of the box, are two equal, opposing through-holes. Each flange is drilled with holes to allow the bracket to be bolted to a roof rack rail. A top bracket made in a box shape with an open bottom to fit over the box-shaped bottom bracket and with the top of the top bracket fabricated to accept two bolts that will screw into an adapter fitted into a rail slot on the underside of a roof-top tent. The top bracket also has two opposing through-holes matching the through-holes on the sides of the bottom bracket. The top and bottom brackets held together by a locking quick-release pin inserted through the through-hole on the sides of the top and bottom brackets.

Inventors: **Tillman-Davis; Michael William (Decatur, GA)**

Applicant: **Tillman-Davis; Michael William** (Decatur, GA)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1751518	12/1929	Maclean	135/147	E04H 15/06
2770244	12/1955	Carson	135/88.13	E04H 15/06
2825351	12/1957	Thornton	135/904	E04H 15/06
2828035	12/1957	Kuchinskie	414/462	B60R 9/042
3192937	12/1964	Carter	135/904	E04H 15/06
3228406	12/1965	Jamba	135/904	E04H 15/06
3289684	12/1965	Lowe	296/159	E04H 15/06
5769482	12/1997	Kirk	296/105	B60J 7/102
5988195	12/1998	Kaestner	135/88.13	E04H 15/06
6192909	12/2000	Strausser	5/113	E04H 15/40
7162968	12/2006	Thompson	114/361	B63B 17/02
2007/0209693	12/2006	Banfill	135/88.08	E04H 15/06
2017/0349080	12/2016	Sautter	N/A	B60P 3/34
2021/0370751	12/2020	Schafer, Jr.	N/A	B60J 7/102

Primary Examiner: Wiley; Daniel J

Attorney, Agent or Firm: Gurr & Brande, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) N/A
BACKGROUND
(2) The inspiration for this invention is to create a rapid-release method to affix a rooftop tent to a vehicle roof-rack and to quickly and easily remove the rooftop tent from the vehicle roof-rack. The space between the bottom of the rooftop tent and the vehicle roof rack is very small . . . no more than 6 inches. Generally a person must use up to eight (8) nuts and bolts to secure a rooftop tent to a vehicle roof rack. This requires several hours pushing ones arms into a cramped, metal space to affix the nuts to the bolts and then several hours uninstalling the nuts and bolts leading to lost time and leaving arms with bruises and abrasions.
SUMMARY OF THE INVENTION
(3) This invention solves the problem of the extensive time and energy lost securing nuts and bolts in the cramped, hard-to-access space between the bottom of a rooftop tent and the top of a vehicle. This invention consists of three parts: part one bolts to a vehicle roof-rack, part two bolts to the underside of a rooftop tent, and part three fits through opposing holes in the sides of the parts one and two after part two has been lowered onto part one. The insertion of part three locks in place

parts one and two thereby securing the rooftop tent to the vehicle roof-rack. The present invention allows the user to insert one piece of locking hardware to secure their rooftop tent to their roof-rack and then to remove that same piece of locking hardware to remove their rooftop tent from their roof-rack.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates a front perspective view of rooftop tent mount apparatus in a disassembled configuration; and

(2) FIG. **2** illustrates a front perspective view of a rooftop tent mount apparatus in an assembled configuration.

DETAILED DESCRIPTION OF THE INVENTION

(3) The purpose of this invention is to serve as a rapid-release/quick-release mount for a rooftop tent.

(4) People wanting to secure a rooftop tent to a vehicle roof-rack have limited options. The most frequently used option is to bolt the rooftop tent to the rack. The following legend provides a numerical guide to the relevant parts of the rooftop tent mount apparatus **100**: bottom bracket **102**; top bracket **104**; locking quick-release pin **106**; flanges **108**A-B of the bottom bracket **102**; opposing holes **110**A-B drilled in the sides of the bottom bracket **102** and opposing holes **112**A-B in the sides of the top bracket **104** where the locking quick-release pin **106** is inserted; holes **114**A-D to bolt bottom bracket **102** to a roof-rack rail; and holes **116**A-B to bolt top bracket **104** to an underside of a rooftop tent.

(5) FIG. **1** shows the bottom bracket **102**, top bracket **104**, and locking quick-release pin **106** unassembled as seen from the front or back, being mirror images of each other. FIG. **2** shows the top bracket **104** and bottom bracket **102** and locking quick-release pin **106** assembled as seen from the front or back, being mirror images of each other.

(6) The rapid-release rooftop tent mount apparatus disclosed herein employs a locking quick-release pin **106** to secure a flanged bottom bracket **102** to a top bracket **104** on the underside of the rooftop tent. The flanged bottom bracket **102** consists of fabricated steel formed into the shape of a box **118** one to two inches high, by one to two inches wide by one to two inches long. The flanges **108**A-B are on opposite sides of the box **118** with each flange **108**A-B having two holes **114**A-B, **114**C-D, respectively, so that the bracket **102** may be bolted to the roof-rack. The box **118** atop and between the flanges **108**A-B has an equal and opposing hole **110**A-B each on opposite sides of the box **118** and above the flanges **108**A-B, respectively. The top bracket **104** is to be bolted to the underside of the rooftop tent. The top bracket **104** is meant to fit over the bottom bracket **102**. The top bracket **104** also consists of fabricated steel but formed into the shape of a lower-case letter “n” with a flat, as opposed to round, top. The underside of the top bracket **104** is left open to fit over the box **118** of the bottom bracket **102**. The top, flat portion **120** of the top bracket **104** has two holes **116**A-B that are used to bolt the top bracket **104** to the underside of the rooftop tent. The top bracket **104** is similarly one to two inches high, by one to two inches wide by one to two inches long and has holes **112**A-B in the side that match the size and location of the side-cut holes **110**A-B on the bottom bracket **102**.

(7) Once the top bracket **104** is secured to the underside of the rooftop tent the apparatus can be lowered on top of the bottom bracket **102** that is bolted to the roof-rack. Where the holes line up on the bottom bracket **102** and top bracket **104**, in other words, hole **112**A of the top bracket **104** aligns with hole **110**A of the bottom bracket **102** and hole **112**B of the top bracket **104** aligns with hole **110**B of the bottom bracket **102**, a locking quick-release pin **106** is inserted, as shown in FIG.

2, to hold the bottom bracket **102** and top bracket **104** together and thereby mount the rooftop tent on the roof-rack.

Claims

1. A rooftop tent mount apparatus, comprising: a bottom bracket configured to secure to a roof-rack of a vehicle, the bottom bracket, comprising: a first, horizontal portion forming a first flange and a second flange, the first flange comprising a plurality of first holes and the second flange comprising a plurality of second holes, the plurality of first and second holes configured to secure the bottom bracket to the roof-rack rail of the vehicle, a box extending vertically upward from the first, horizontal portion and interposed between the plurality of first holes and the plurality of second holes of the first and second flanges, respectively, the box comprising a first wall having a first hole and a second wall having a second, opposing hole; a top bracket, comprising a first, horizontal portion having a first hole on a first end and a second hole on a second end, a second, vertical portion extending downwardly from the first, horizontal portion and interposed between the first and second holes of the first, horizontal portion, the second, vertical portion comprising a first wall having a first hole and a second wall having a second, opposing hole; and a locking pin; wherein the second, vertical portion of the top bracket is configured to slide over and couple to the box of the bottom bracket, the first hole of the first wall of the second, vertical portion of the top bracket aligning with the first hole of the first wall of the box of the bottom bracket, and the second hole of the second wall of the second, vertical portion of the top bracket aligning with the second hole of the second wall of the box of the bottom bracket; wherein when the top bracket is coupled to the bottom bracket, the locking pin is configured to slide through the first hole of the of the first wall of the second, vertical portion of the top bracket, through the first hole of the first wall of the box of the bottom bracket, through the second hole of the second wall of the box of the bottom bracket, and through the second hole of the second wall of the second, vertical portion of the top bracket, thereby preventing separation of the top bracket from the bottom bracket; and wherein when the top bracket is coupled to the bottom bracket, the first, horizontal portion of the top bracket is perpendicular to the first, horizontal portion of the bottom bracket.